

Think, Grab, and Play: Learning Mathematics in Elementary School with Manipulatives and GCompris, Supported by Computational Thinking

ABSTRACT

This paper presents the results of an action research aimed at evaluating a method to stimulate the learning of mathematics with the support of computational thinking in 1st grade elementary school classes. The study was conducted in a public school, involving children aged between 6 and 8 years old, and comprises the research stages defined as: planning, mediation of activities, observation, and reflection. The pedagogical proposal integrates the strategy of using Montessori Golden Beads and educational software as resources to explore basic mathematical concepts in a playful and interactive way. Data collection included classroom observations, records through videos and photographs, reports from the teacher and notes in a field diary, analyzed through content analysis. The results indicate that the combined use of physical and digital resources favored the children's involvement and expanded their understanding of concepts such as quantity, addition, grouping and association. The research contributes evidence on the potential of computational thinking as a strategy to support mathematical learning in the early years.

Author Keywords

Computational Thinking; GCompris; Hands-on Manipulatives for Math; Instructional Approach

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•Applied computing~Education~Computer-assisted instruction

INTRODUCTION

Despite constant efforts in previous research [9, 20, 22], students still face clear difficulties in studying and understanding Mathematics. Data on Brazilian students' performance, both in comparison to the expected national standards (such as IDEB and SAEB¹) and to international benchmarks (according to PISA² – Programme for International Student Assessment), reflect this issue.

Teaching Mathematics in the early years of Elementary School presents recurring challenges, particularly regarding the development of mental arithmetic for addition, the recognition of multiple values, and the understanding of place value. These skills are essential for mathematical literacy and for students' progression to more complex topics [3]. However, even though Brazil's National Common Curricular Base (BNCC, in Portuguese *Base Nacional Comum Curricular*) emphasizes the use of active methodologies in the classroom [3], along with manipulatives and digital technologies, traditional approaches that remain disconnected from students' daily lives still prevail [18].

In this context, this paper presents and analyzes a methodological proposal developed through action research in a first-grade class. The proposed approach, *Think, Grab, and Play*, combines Golden Bricks manipulatives with the GCompris educational software as a strategy to strengthen and enrich the learning of mathematical concepts. It is grounded in Computational Thinking (CT) [24] and in the general competencies of the BNCC, applying both manipulatives and digital resources. The main goal of this approach is to foster students' interest in Mathematics, increase participation and engagement, and stimulate curiosity toward problem solving and calculation. This study was conducted through an action research approach, with the participating researcher acting as an

¹<https://todospelaeducacao.org.br/noticias/ideb-e-saeb-veja-os-destaques-dos-resultados-de-2023/>

²<https://www.gov.br/inep/pt-br/assuntos/noticias/acoes-internacionais/divulgados-os-resultados-do-pisa-2022>

Decomposition	Abstraction	Pattern Recognition	Algorithmic Design
It breaks down a complex problem into smaller, more manageable parts that are easier to solve.	It ignores irrelevant details and focuses on the most important aspects of a problem.	It identifies similarities, regularities, or trends that help in understanding and solving problems.	It creates a logical, well-defined sequence of steps for solving a problem.

Table 1. Computational Thinking Pillars.

outsider [11]. The head teacher contributed by supervising and reporting her perceptions, while three additional researchers implemented the approach in a public elementary school. Data collection involved action research journals, audiovisual recordings, notes from message groups, and continuous participant feedback, with the goal of documenting and reflecting on the ongoing process.

The pedagogical approach “*Think, Grab, and Play*” is based on respecting screen time during the use of educational games [21], as well as on understanding the attention span that each child can sustain in an activity. Therefore, the methodology includes transitions every 20 minutes, totaling one hour of intervention per session.

The results, after 13 in-person sessions focused on the introduction and use of manipulative material and educational games, showed that the approach was well accepted by the students. In addition, we observed progress in the development of logical reasoning and mental calculation, evidenced through written assessments and questions posed by the head teacher. On the other hand, given the difficulties identified and the differences in learning pace, it was necessary to repeat the use of some games and the algorithm required for working with manipulatives, which demanded more time than initially planned.

This article is organized as follows: the *Background* section presents the foundation for understanding the study, followed by the *Related Work* section. The *Methodological Path* section details the context and the action research approach. The section dedicated to the “*Think, Grab, and Play*” approach presents it as a strategy for teaching mathematics through concrete and digital resources, supported by Computational Thinking. The subsequent section describes the practical application of this pedagogical approach carried out in a public elementary school. The *Results and Discussion* section discusses the main findings and presents the lessons learned. Finally, the *Conclusions and Future Work* section provides some preliminary conclusions and points out directions for further research.

BACKGROUND

The foundational theories of this study are related to Computational Thinking as a broader teaching and learning method, the use of manipulatives in early childhood education for the appropriation of abstract mathematical concepts, and the GCompris software as a supportive

learning resource. These elements are understood as complementary, contributing to a more inclusive learning process capable of addressing the educational needs of students with different profiles. This section presents these topics in greater detail.

Computational Thinking

Computational Thinking (CT) has its roots in teaching programming to children, originating from Seymour Papert’s work in the 1980s [17]. However, it was only in the 2000s that the term gained popularity [24]. The central idea of CT is to enable the resolution of any problem, using its four pillars as a foundation to systematize solutions.

The four pillars of CT, summarized in Table 1, correspond to cognitive skills that are transversal to any problem, allowing the concept to go beyond the boundaries of computing itself and be applied in different contexts. **Abstraction** reduces the complexity of a problem’s elements by emphasizing only what is essential for its solution [24]. **Decomposition** enables a larger problem to be divided into smaller, independently solvable problems, whose combined solutions address the overall challenge [24]. **Pattern recognition** makes it possible to generalize solutions by identifying common features in data, as well as by establishing connections and similarities that can be applied in other contexts [24]. Finally, understanding what an **algorithm** is allows for the definition of clear steps to solve a specific problem, creating sequences of instructions that guide the necessary decisions or repetitions for execution [24].

In this sense, CT, in addition to being a transversal competence applicable to different areas of knowledge, is directly aligned with Educational Informatics. This field not only seeks to introduce digital technologies into the school context but also to employ them pedagogically, expanding teaching and learning possibilities [1, 4]. Thus, the development of CT in children can be understood as a fundamental dimension of Educational Informatics, as it fosters cognitive skills, autonomy, and problem-solving strategies supported by digital resources [1].

Manipulatives in Early Childhood Education

The use of concrete materials or manipulatives in early childhood education dates back to when Maria Montessori, a physician and educator, created materials to support learners with learning difficulties in making mathematics more tangible, particularly with respect to the decimal number

system [16]. In this context, the Golden Bricks (*material dourado*) were developed to assist in this task.

This pedagogical resource provides a visual and tactile representation of quantities in units, tens, hundreds, and thousands. By associating quantities through the use of Golden Bricks, children can count and understand the value of each digit, thereby grasping the decimal system. The intentional use of Golden Bricks facilitates the comprehension of the four basic operations, beginning with mathematical facts and progressing toward more complex calculations [16]. Such activities can foster logical reasoning and abstraction skills while also encouraging children's independence in performing basic operations, making mathematics both concrete and engaging [16].

Another manipulative that deserves attention is the abacus. According to many historians, the abacus is a calculating instrument that dates back 5,000 years. Several variations of this tool exist, such as the Japanese *soroban*, the Chinese *suapan*, the Cranmer abacus (designed for the visually impaired), and the open-frame abacus, the most common type in schools. Compared to the Golden Bricks, it involves a higher level of abstraction but provides important insights into the decimal number system, particularly positional value [12].

Among the benefits of using the abacus in early childhood education are enhanced mental calculation agility, logical reasoning, and concentration [12].

The GCompris Educational Software Suite

GCompris is an educational software suite designed for teaching various subjects, including mathematics, through programs that are easy to understand for both children and teachers. It is available in more than 50 languages and is targeted at children aged 2 to 10 years [6]. Created in 2000 as open-source software, it is distributed under the GNU General Public License. Free versions are available for Linux and macOS; however, its use on Windows may require the purchase of a paid license.

RELATED WORK

Starting from abstract concepts and understanding them through concrete objects is an effective strategy in Mathematics education, with the potential to engage students, including those with learning difficulties, as demonstrated in previous studies [2, 13]. These related works proposed the use of manipulatives to address this challenge. However, they differ from the approach presented in this paper, as they do not integrate concrete materials with the use of digital games.

Other studies have used the concept of Golden Bricks (*material dourado*) as the basis for developing educational software that simulates its manipulation [2, 7, 20, 22]. This work, in contrast, proposes the use of Golden Bricks as a complementary concrete resource, combined with other

physical support tools such as the ten frame, the hundred chart, and the abacus.

Finally, studies that explore Computational Thinking examine its relationship with Mathematics and the BNCC [25], discuss its development in early elementary students [9], and analyze how its pillars can be integrated into methodologies such as problem-based learning [14]. In contrast, this paper proposes the use of Computational Thinking as a strategy to systematize mathematical learning, combining it with manipulatives in the teaching process.

METHODOLOGICAL PATH

The methodology adopted in this research, qualitative and applied in nature, is characterized by participatory investigation, with broad and explicit interaction between the researcher and the actors involved in the real situation. It employs the action research method through the design of mathematics activities based on the expected skills outlined in the BNCC, focusing on addition and place value. In this context, the four pillars of Computational Thinking were applied both to problem-solving and as support for mental calculation, using Golden Bricks and the GCompris software suite as complementary resources.

With the purpose of constituting a planned action in practice, the choice of action research stands out, as it enables reflection on practice while simultaneously seeking to clarify a real situation in each social context, thereby expanding knowledge. The action research cycle adopted included the stages of Planning, Acting, Observing, and Reflecting, as presented in Figure 1.



Figure 1. Stages of the action research performed in this work.

Planning the collective action began with the diagnosis and identification of the problem situation in a public school located in an urban area, but which also serves a rural community due to its proximity to the countryside. The target audience of the intervention consisted of first-grade students in Elementary School, with an average age of 6 years, enrolled in a municipal school situated in a lower-middle-class neighborhood. The school serves both students from

Hands-on Manipulative	Decomposition	Abstraction	Pattern Recognition	Algorithm Design
Ten Frames	Visual representation of numbers as smaller parts (e.g., 7 as $5 + 2$), facilitating understanding of operations.	Use of paper circles to represent numerical quantities.	Identify arrangements of circles that represent numbers.	Perform sequential steps to solve additions and subtractions on the ten frames in a structured way.
Golden Bricks	Breaking numbers into smaller components (e.g., 134 as $100 + 30 + 4$) to facilitate operations and analysis.	Representation of units, tens, and hundreds using cubes, rods, and flats, simplifying understanding of place value.	Observation of regularities in the decimal system, such as the repetition of groups of 10 to form tens and hundreds.	Application of rules and ordered steps to perform additions, subtractions, and exchanges between units, tens, and hundreds.
Open Abacus	Separation of numbers by place value (e.g., 253 as $200 + 50 + 3$), facilitating visualization and understanding of place value.	Representation of numerical values using rods with markers, highlighting only essential information (units, tens, hundreds).	Identification of regularities in the grouping of values in each place of the decimal system.	Performing mathematical operations through sequential movements of markers, following defined rules (e.g., exchanging 10 units for 1 ten).
Activity Notebook	When solving an activity, the student breaks the problem into steps (read, understand, solve, record the answer).	The child needs to focus only on the relevant information in each task (e.g., numbers, shapes, colors), ignoring distractions.	By repeating similar activities, the student identifies patterns in the instructions, numerical sequences, or figures.	Follow step-by-step instructions to complete activities (e.g., counting by 2s, coloring according to the model, connecting the dots).

Table 2. Relationship of Unplugged Activities with the Pillars of Computational Thinking.

the local neighborhood and those from rural areas who are in situations of vulnerability.

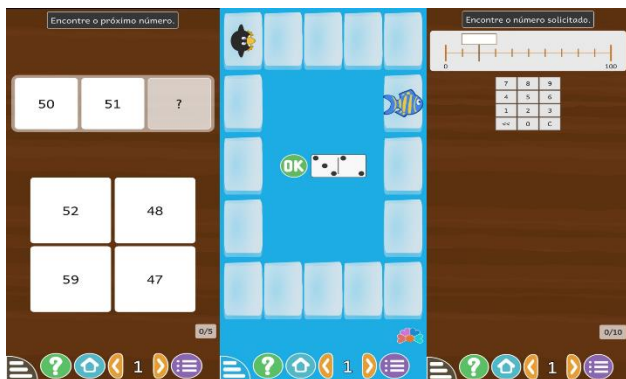


Figure 2. Examples of three games on GCompris (Find the adjacent numbers, Count intervals and Read a graduated line).

The activities were conducted in the afternoon, with a class of 16 students, including 9 girls and 7 boys. All the didactic material used was individual, including base-ten blocks, an open abacus, and Chromebooks. The Chromebooks allowed interaction with the GCompris software through a

touchscreen interface, featuring interactive games in Portuguese. At this stage, we selected manipulatives to be used in each intervention, as well as choosing educational games within GCompris to reinforce skills. Table 2 presents the description of the games used in the interventions, and Figure 2 shows screenshots of three of these games.

In the *Acting* stage, the planning from the previous phase was implemented through weekly interventions. As planned, each session lasted one hour, divided into three 20-minute transitions. Each transition was associated with a verb from the pedagogical approach *Think, Grab, and Play* presented in this article. We dedicated the first 20 to presenting the theoretical content of the lesson and the use of base-ten blocks associated with problem-solving (*Think*). The following 20 minutes focused on using base-ten blocks or the 100-board to solve the proposed exercise (*Grab*). The final 20 minutes were dedicated to plugged-in activities with GCompris educational software (*Play*), which aligns with recommended practices [21]. Moreover, the division into 20-minute activity blocks considered children's attention span in this age group, which ranges from 14 to 35 minutes. In total, we held 13 sessions between July and November 2024.

During the activities, the participant-researcher, along with the other participants, forming a team of four people, applied and supervised the activities, collected student feedback, observed their outcomes, and recorded them in photos and videos. After each session, a logbook was prepared with notes on student results and difficulties during this stage, thus covering the *Observing* stage.

Finally, in the *Reflecting* stage, the action-research cycle was consolidated, though not necessarily concluded. The iterative nature of the applied method was confirmed in the cyclical process of diagnosing a problem, acting, observing, and critically reflecting on the results achieved through the analysis of strategic points, mobilization capacity, quality of teamwork, communication skills, and continuity of the project [11].

In the case under analysis, the activities carried out and the results obtained were verified based on materials derived from observation, photos, videos, and reports from the lead teacher, duly recorded in the logbook. Through the learning and insights gained from the experience, the method suggests returning to the starting point to highlight the generalizable knowledge acquired about the studied matter. Thus, the participant-researcher conducted an analysis of the results, and the deliberations with the participants pointed toward the continuation of the study and the evolution of the action plan for a subsequent intervention, considering the iterative nature of action research [11].

The intervention was developed from the identification of a recurring problem by the lead teacher of the first-grade class in Elementary School: the limited agility in performing mental calculations and logical reasoning, combined with difficulties in solving addition operations and recognizing numerical patterns.

Accordingly, the proposed approach seeks to align with the concept of Computational Thinking, applying its pillars—as presented in Table 1—in association with the proposed games and the problem-solving methodology.

"THINK, GRAB, AND PLAY": AN APPROACH TO MATHEMATICS IN ELEMENTARY EDUCATION

The "*Think, Grab, and Play*" approach is a pedagogical proposal developed during the action research described in this study for teaching mathematics in the early years of Elementary School. Inspired by constructivist principles and the pillars of Computational Thinking, this approach integrates playful practices with concrete and digital resources to foster the development of logical-mathematical reasoning.

It was developed based on the identification of a recurring problem by the lead teacher of the first-grade class: limited agility in mental calculation and logical reasoning, combined with difficulties in performing addition operations and recognizing numerical patterns. Accordingly, the proposed

approach seeks to align with the concept of Computational Thinking, applying its pillars in conjunction with the proposed games and the problem-solving methodology.

Figure 3 presents an overview of the proposed approach, which is structured around three complementary pillars:

- **Think:** Students are invited to reflect on the proposed mathematical challenge, anticipating solution strategies based on prior experiences, hypotheses, and dialogue. This phase emphasizes metacognition and informal mathematical reasoning [19], stimulating abstraction, decomposition, pattern analysis, and planning—core characteristics of Computational Thinking [24].
- **Grab:** In this stage, students manipulate concrete materials such as Golden Bricks to represent, test, and adjust their ideas. *Grab* emphasizes free play, rule-setting, and diverse representations, together with sensorimotor engagement and the use of physical representations to mediate knowledge construction [10, 13].
- **Play:** Learning is consolidated through interaction with digital games that contextualize mathematical concepts in problem situations. The games promote engagement and collaboration while reinforcing skills [5].

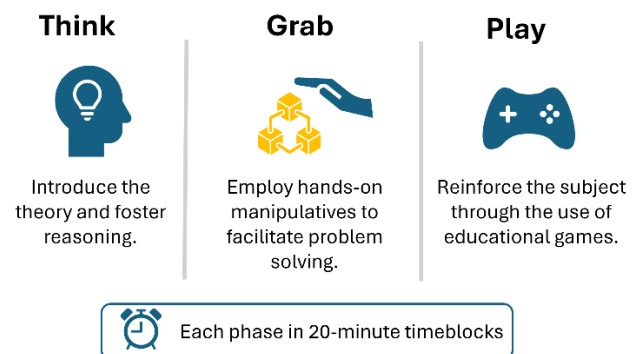


Figure 3. Think, Grab, and Play Approach.

In addition to these pillars, the approach aligns with the BNCC, working on foundational skills and expanding as students make progress. By linking the concrete and the abstract to understand mathematical abstractions, students are expected to experience mathematics in an active, creative, and meaningful way, integrating theory and practice, concrete representation, and progressive abstraction. The approach also allows the teacher to act as a mediator of the process, adapting challenges to the class's needs and fostering student autonomy. approach

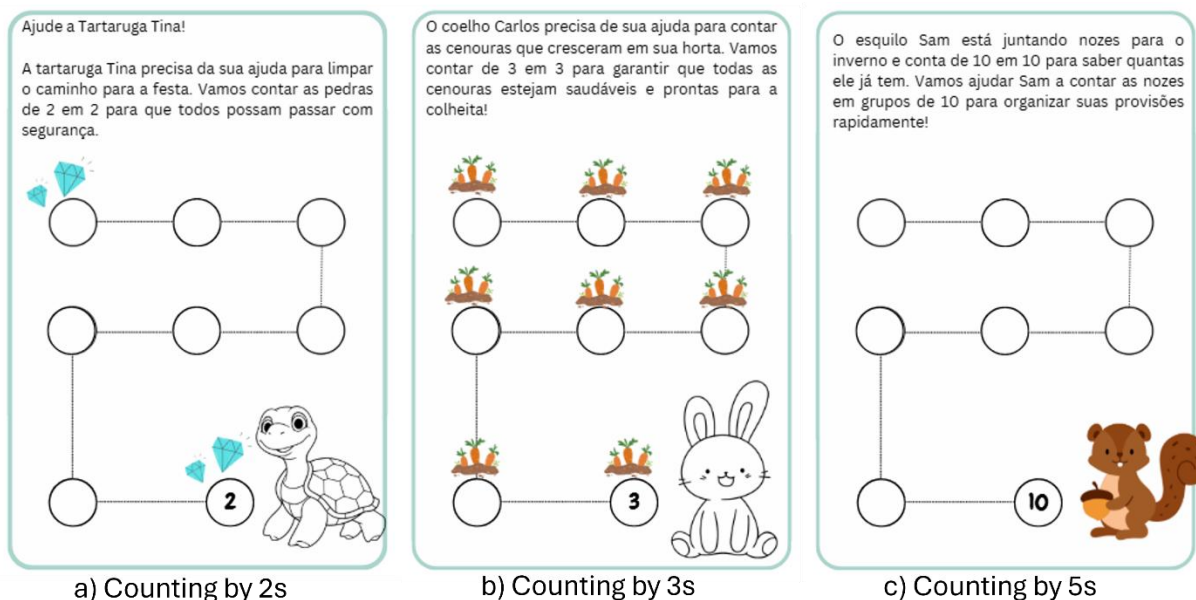


Figure 4. Three unplugged activities conducted with the children.

The *Think, Grab, and Play* approach further supports the development of students' metacognitive skills by encouraging reflection on the strategies used to solve problems, monitoring their own actions during problem-solving, and evaluating their results. This process is particularly relevant in the context of mathematics learning, as it contributes to the development of student autonomy and self-confidence.

APPLICATION OF THE "THINK, GRAB, AND PLAY" PEDAGOGICAL APPROACH

In the context of teaching and learning, Educational Informatics focuses on the study and pedagogical application of digital technologies, expanding the use of software to support knowledge construction and innovation in educational practices [8, 23]. In this sense, the proposed in this study constitutes an application of Educational Informatics by integrating the conscious and planned use of digital educational games and concrete materials, providing alternative and complementary strategies aimed at addressing the diverse needs and learning styles of students. Considering the difficulties many students face in learning mathematics, as well as their familiarity with technological devices, the pedagogical intervention was planned to focus not only on "thinking" but also on "building," as highlighted by [15] regarding Seymour Papert's proposal. According to the authors, constructing concrete representations of abstract mathematical concepts, mediated by computational resources, enables students to develop their own mental models for understanding the world. Thus, the intervention sought to explore not only the pillars of Computational Thinking but also to foster student autonomy in the proposed activities.

The sessions lasted approximately one hour each, organized according to the *Think, Handle, and Play*

approach, in three segments of 20 minutes each. The first segment was dedicated to playful activities introducing and reinforcing mathematical concepts through the pillars of Computational Thinking. In the second segment, students participated in practical activities with manipulatives, a phase aimed at free exploration, familiarization, and application of abstraction, decomposition, and pattern recognition concepts. Finally, the third segment took place in the computer lab using the GCompris suite, where students reinforced and applied concepts in a relaxed and interactive environment. As the school where the activities were conducted had a lab equipped with Linux laptops, the use of GCompris was feasible.

The first segment of each session, marked by theoretical introduction and contextualization of concepts, engaged students through short stories presented in an activity notebook designed for the intervention. In this notebook, the proposed activity was explained with the aid of a whiteboard and marker. The manipulatives to be explored later were also introduced, along with the algorithm for their use, reinforcing the concept of composing and decomposing tens. This initial segment was essential, as it required greater student attention and focus, allowing prior assimilation of the content to be deepened in subsequent stages.

In the second, practical and concrete segment, students were able to apply the concepts previously worked on, using Golden Bricks and the open-frame abacus as primary resources. This stage was crucial for bridging theory and practice, supporting the construction of mathematical knowledge through problem-solving.

Figure 4 shows unplugged activities carried out on three different days with the students, using Golden Bricks.

Activity a) *Counting by twos* was students' first contact with Golden Bricks and was easy to understand; however, the concept of composition and decomposition proved more challenging. Activities b) and c) in Figure 4 were important for students to appropriate these concepts.

The third segment focused on applying the pillars of Computational Thinking through digital games. In this stage, students developed autonomy by interacting with the virtual environment, respecting screen-time limits, and exploring the computational space meaningfully. In each session, one GCompris game was used, providing an alternative approach to counting, simple addition, and subtraction.

For the study presented in this article, the following BNCC skills [3] were selected, aligned with the difficulties identified by the lead teacher of the first-grade class:

- **EF01MA02:** Count accurately or approximately, using different strategies such as pairing and other groupings.
- **EF01MA04:** Count the number of objects in collections up to 100 units and present the result through verbal and symbolic records, in situations of personal interest, such as games, play activities, classroom materials, among others.
- **EF01MA05:** Compare and order natural numbers up to two digits, with or without support from the number line.
- **EF01MA07:** Compose and decompose numbers up to two orders, through different additions, with the support of manipulatives, contributing to the understanding of the characteristics of the decimal number system and the development of calculation strategies.
- **EF01MA08:** Solve and formulate addition and subtraction problems involving numbers up to two digits, considering the meanings of joining, adding, separating, and removing, with the support of images and/or manipulatives, using personal strategies and recording methods.

At the end of each session, discussions were held with the lead teacher to assess students' understanding, identify their needs, and adjust activities as necessary. This reflective process was of utmost importance for the continuous improvement of the intervention and to ensure its pedagogical effectiveness.

Implementation of the Activities

The activities were planned to integrate concrete and digital resources, illustrating mathematical concepts in diverse ways for the students. After presenting the content, unplugged activities were carried out using manipulatives such as Golden Bricks and the open-frame abacus. Suggested unplugged activities are illustrated in Figure 4. To support these activities, the manipulatives listed in Table 2 were used. Subsequently, students engaged with

games from the GCompris software. A total of eight educational games addressing the same content in an interactive format were used.

The activities took place over 13 in-person sessions, each lasting approximately one hour, conducted between July and November 2024. The first session had an introductory nature, with a playful and welcoming approach. During this session, students were introduced to the intervention team, which included one researcher and three collaborators. This initial moment was essential to ensure the active participation of all students, promoting an inclusive and stimulating learning environment.

The objectives of the proposal were presented in an accessible and interactive manner, highlighting the relevance of mathematics and its connection to technology. Emphasis was placed on how this knowledge is essential throughout life and how computers are part of students' daily routines. The activity developed on that day was unplugged, using the "Ten Frame." Each student received a sheet with two columns of five squares, along with ten paper circles serving as counters. The objective was to represent numbers from 0 to 10 using the counters, also exploring simple addition and subtraction operations. This initial activity showed that students understood the abstraction of numbers from one to ten. During the activity, in addition to mathematical content, the pillars of Computational Thinking were also addressed.

In the second session, the Golden Bricks were introduced, and their use continued for eight more sessions, supported by the pillars of Computational Thinking to reinforce learning, as shown in Table 2. This resource provided a concrete representation of numbers and their groupings, focusing first on counting units and then gradually transitioning to tens and hundreds.

Manipulating the blocks allowed students to experience practical counting and numerical organization, promoting a stronger understanding of grouping logic and reinforcing the relationship between physical manipulation and the symbolic representation of numbers. To complement the concepts worked on, students were guided to the computer lab to play the game *Count the Items*, where they developed basic counting skills and associations between quantity and number.

In the third session, the game *Count Intervals* focused on developing sequential counting while intuitively introducing concepts related to multiplication through abstraction. Using Golden Bricks, students initially worked on pattern identification by grouping in twos. In the fourth session, counting by threes was explored. Additionally, based on the lead teacher's observation of students' difficulty writing some numbers in a mirrored fashion, the game *Draw Numbers* was used. This helped students recognize the visual structure of numerals, working on fine motor coordination and understanding that different visual forms (writing) represent the same numerical concept.

In the fifth session, using Golden Bricks, students grouped numbers by fours, performed addition, and were introduced to the concept of regrouping tens. The game *Learn Digits* was applied playfully to support recognition and association between numbers and quantities. While interacting with the game, students identified regularities in numerical representation (digits) and their corresponding quantities, fostering abstraction with Golden Bricks.

The sixth session focused on the spatial location and relationships between numbers, where students had to identify which numbers were positioned around a given number. To consolidate this concept, the game *Find the Adjacent Numbers* was used. Notably, from this session onward, students were provided with an individual 100-frame to use as a reference in all activities.

The seventh session introduced a higher level of complexity compared to previous sessions, as concepts were presented. Initially, the principles of grouping were reinforced using Golden Bricks. Then, students transitioned from unit-based quantity concepts to representation in tens. They grouped cubes in eights, linking this practice to addition operations, and gradually progressed to grouping by tens, reinforcing the mechanics of regrouping. This process enabled the concrete introduction of the concept of ten, connecting it to the tens columns in Golden Bricks.

The main objective of this session was to automate counting through composition in tens via regrouping and to identify numerical patterns emerging from counting by tens, encouraging students to represent numbers using both units and tens. With progressive mastery of Golden Bricks, students developed a more structured understanding of numerical composition. In the computer lab, the digital activity was the game *Use the graduated line*, which reinforced the concept of tens and place value.

In the eighth, ninth, and tenth sessions, the approach focused on not introducing new concepts but reinforcing the transition from units to tens, strengthening place value understanding, addition skills, and logical reasoning. The digital activities involved the games *Count Intervals* and *Frieze*.

Introduction to the open-frame abacus began in the eleventh session and continued until the end of the intervention, allowing students to explore the material freely and interact with the beads, promoting initial familiarization in a playful manner. Then, the rules for using the abacus were introduced, explaining that each rod represents a numerical order (units, tens, hundreds) and that beads should be organized from left to right, respecting place value. Students were instructed that each bead represents one decimal unit and that upon reaching ten beads on a rod, they should be exchanged for the next order. Additionally, a rule was established that each rod

should contain beads of only one color, facilitating visualization and association between numerical orders and their groupings.

In the computer lab, the concepts of the open-frame abacus were reinforced using the game *Read a graduated line*, in which a partially visible number line was presented, and students identified the number corresponding to the position of a marker based on the displayed sequence. This activity strengthened logical reasoning, forward and backward counting, spatial number location, and reinforced the pillars of Computational Thinking. Table 3 presents the observed relationship between the games used and Computational Thinking before and during their application.

RESULTS AND DISCUSSION

The students' reception to the activities was very positive, which allowed for the stimulation of both perception and logical construction through proposals integrated with Computational Thinking using the GCompris software and concrete materials commonly found in the classroom, such as Golden Bricks and the open-frame abacus. In the first sessions, some students experienced difficulties with grouped counting, particularly because it was the first experience for many with Golden Bricks.

The interactive interface of the digital games contributed to maintaining student engagement, supporting the assimilation of content in a playful and meaningful way. From the fourth session onward, students became more familiar with the activity dynamics, completing tasks with greater agility and spontaneity.

Many students showed interest in revisiting previously worked-on games, which demonstrated not only a growing mastery of the digital interface but also significant engagement with the content covered. This behavior served as an important indicator of learning and provided positive feedback to the intervention team.

Although some Computational Thinking concepts are abstract for the targeted age group, practicing with games allowed for an accessible introduction. For example, when searching for numbers around a central value, students applied simple rules, such as subtracting 1 to locate the number on the left, informally executing an algorithm without full awareness. This approach supported the development of logical reasoning, attention, and problem-solving skills, as well as helping in the understanding of spatial and numerical structures.

During the transition from units to tens and hundreds, it was observed that some students represented numbers like 13 using only units (13 loose cubes), without grouping them into one ten and three units. The classroom teacher played a key role in identifying these difficulties and reinforcing the concepts over the days, which contributed to consolidating learning. new

Name of the Game	Decomposition	Abstraction	Pattern Recognition	Algorithm Design
Count the items	Focus on counting one group at a time, breaking the task into smaller parts.	Focus on the number of repetitions, ignoring irrelevant details.	Identifies patterns in the arrangement and repetition of objects, facilitating counting.	Develops a mental process to count in an organized way, creating a logical sequence of actions.
Count intervals	Step-by-step analysis of the numerical sequence to identify missing numbers	Focuses on the counting rule, ignoring numbers that do not follow the pattern.	Identifies the logic of the sequence (e.g., +2, +5) to predict the next numbers.	Applies a clear and repetitive rule (e.g., 'add 3') to correctly complete the sequence.
Draw numbers	It is necessary to follow the tracing number by number, breaking the process into smaller steps.	Understanding that different visual forms represent the same numerical concept.	Observes the shape of numbers and identifies patterns in the strokes to facilitate writing the next ones.	Follows a sequence of defined steps to correctly draw each number, as a procedure to be repeated.
Learn digits	Exercise one digit at a time, breaking the learning process into small parts.	Focus on the essential elements of the number (shape and sound), ignoring visual variations or different contexts.	Identify visual similarities between digits and repeated associations between sound and symbol.	Create a mental routine to systematically recognize, pronounce, and memorize each digit.
Find the adjacent numbers	Break the task into parts, such as identifying the target number and moving along the number line to locate it.	Focus on the values and the distances between them, ignoring irrelevant visual elements such as color or style.	Identify the spacing pattern between numbers on the number line (e.g., +1, +2, +5, etc.).	Develop a logical strategy to position numbers based on their relationship with others (e.g., always locate the extremes first, then the middle).
Use a graduate line	The child breaks the task into parts, such as identifying the target number and moving along the number line to locate it.	Focuses on the values and the distances between them, ignoring irrelevant visual elements such as color or style.	Identifies the spacing pattern between numbers on the number line (e.g., +1, +2, +5, etc.).	Develops a logical strategy to position numbers based on their relationship with others (e.g., always locate the extremes first, then the middle).
Frieze	The child analyzes the pattern by separating each element (color, shape, and position) to understand its structure.	Ignore irrelevant details and focuses only on the essential attributes (shape and color) that make up the pattern.	Identify the repetitive visual sequence with shapes, colors, and positions, using this pattern to predict the next elements.	Observation and identification of a pattern that is pre-established and provided by the game for solving a problem.
Read a graduated line	The child breaks the task into steps: identify the number, locate the corresponding position, and drag it to the correct place.	The child understands that the numbers on the number line represent relative positions rather than concrete objects.	Recognize patterns of increment or decrement on the number line (e.g., counting by 1s, 2s, etc.).	Develops a logical strategy to position numbers based on their relationship with others (e.g., always locate the extremes first, then the middle).

Table 3. Relation of GCompris Games with the Pillars of Computational Thinking

In collaboration with the teacher, it was decided that the content presented by the intervention team would be revisited in regular classes, focusing on the areas that presented the greatest challenge.

Some students were able to associate transitions between the Golden Bricks and the open-frame abacus, while others still struggled to understand exchanges between decimal places. As a mediation strategy, integrating these materials

was suggested, promoting a gradual transition between numerical representation systems and supporting the understanding of the concepts involved. Through spontaneous reports to the research-action team, the 1st-grade teacher revealed that the intervention helped students assimilate curricular content, especially in addition and subtraction operations, and promoted the development of logical thinking and student autonomy. The potential of the Golden Bricks was also highlighted, as they began to be used more intentionally and meaningfully in the classroom after the experience. Additionally, the teacher noted that the intervention supported her teaching practice, providing a repertoire for classroom activities and making content presentation more dynamic.

Regarding students who had already exhibited difficulties and required more support, they continued to be insecure. In this report, the teacher identified the need to reinforce the use of concrete materials, promoting a more direct transition to written numbers and addition and subtraction algorithms. The use of GCompris as an additional resource was still insufficient, possibly indicating the need to develop specific games for students who continue to have learning gaps.

Lessons Learned

The implementation of the “Think, Grab, and Play” pedagogical approach in the context of 1st grade in Elementary School revealed important insights for mathematics teaching, namely:

- **Concreteness supports abstraction.** The use of manipulatives such as the Golden Bricks, the Ten Frame, and the abacus reinforced the understanding of concepts like place value, mental calculation, and multiples. By manipulating the objects, students were able to visualize and construct meanings that served as a foundation for faster mental operations.
- **Educational games engage students and reinforce content.** The games from the GCompris suite functioned as an extension of the concrete activities, promoting an environment for experimentation and reinforcement of the concepts studied. The presence of the games increased interest in mathematics, even among students with greater attention or learning difficulties.
- **The involvement of the classroom teacher is essential.** The active collaboration of the class teacher was decisive for the success of the approach. Her mediation and reinforcement, by repeating several of the activities conducted during the intervention, facilitated the integration of the resources into the classroom routine, helping students transfer knowledge across different contexts.

- **Computational Thinking can serve as a learning resource for early-grade students.** Although the students participating in the intervention described in this study were beginning their literacy process, they demonstrated behaviors associated with Computational Thinking, such as decomposing tasks, following logical steps, identifying patterns, and reviewing strategies. This indicates that, when properly adapted, CT can be promoted even in the first years of schooling.

These lessons suggest that the combination of a concrete approach, digital support, teacher mediation, and the principles of Computational Thinking can represent a powerful strategy for teaching mathematics in the early grades, contributing to the development of solid foundations in logical reasoning, autonomy, and critical thinking skills.

CONCLUDING REMARKS

This study presented a pedagogical proposal to support the teaching of mathematics for 1st-grade students in Elementary School. The “Think, Grab, and Play” approach aims to facilitate the transition between the challenging abstraction of calculation and concrete thinking through the use of manipulable materials, such as the Golden Bricks, and the playfulness offered by educational games from the GCompris suite.

The results indicated that the approach was well received by the students and by the school management team. The participation of the classroom teacher was fundamental in reinforcing the concepts covered and integrating the proposed pedagogical resources.

At the end of the intervention, it was observed that some students were still developing the targeted skills, while others met or even exceeded the expectations of the approach. No student performed below expectations, considering the objectives of the “Think, Grab, and Play” approach.

As a next step, the research-action team will return to the same class, now in the 2nd grade of Elementary School, with the aim of expanding the approach in a longitudinal study, monitoring its effects on student performance over time. Additionally, a training session on the approach was requested, motivating the development of support materials aimed at all classes in the early years of Elementary School. An open class was also suggested to present the approach to families, encouraging the continuation of the practices at home.

Considering the classroom teacher’s reports, another potential outcome of this study is to investigate the development and application of digital games specifically designed for students who present persistent difficulties in mathematics, analyzing how these resources can complement the use of concrete materials and support the

transition to symbolic representation and addition and subtraction algorithms.

REFERENCES

1. Acevedo-Borrega, J., Valverde-Berrocso, J., & Garrido-Arroyo, M. D. C. (2022). Computational thinking and educational technology: A scoping review of the literature. *Education Sciences*, 12(1), 39.
2. Alves, A. G., Chaves, J. E., and Cordeiro, A. M. (2020). Interface tangível com material dourado em jogo digital de aprendizagem de matemática. In *Simpósio Brasileiro de Informática na Educação (SBIE)*, pages 612–621. SBC.
3. BRASIL (2018). Base Nacional Comum Curricular. Ministério da Educação, Brasília.
4. Carvalho, F., & Braga, M. (2022). Pensamento Computacional na Educação Brasileira: um olhar segundo artigos do Congresso Brasileiro de Informática na Educação. *Revista Brasileira de Informática na Educação*, 30, 237-261.
5. da Silva, F. N. and da Silva, M. A. (2023). Recursos digitais no componente curricular Matemática: o que dizem os professores dos anos iniciais do ensino fundamental. In *Congresso sobre Tecnologias na Educação (Ctrl+ e)*, pages 71–80. SBC.
6. da Silva, G. Q., Silva, A. M. L., da Silva Sousa, F. A., Aita, K. M. D. S. U., Lima, F. R., & Machado, V. P. (2021). Uso da Ferramenta Computacional GCompris para Auxílio ao Processo de Ensino Aprendizagem do Ensino Fundamental. In *Encontro Unificado de Computação do Piauí (ENUCOMPI)* (pp. 73-80). SBC.
7. da Silva, I. C. S. (2023). Gamificando o material dourado: Uma estratégia para o auxílio do ensino do sistema decimal de numeração para crianças com tea. In *Simpósio Brasileiro de Jogos e Entretenimento Digital (SBGames)*, pages 1397–1402. SBC.
8. de Almeida, M. E. B. (2000). *Informática e formação de professores*. SEED/MEC.
9. de Oliveira, M. C., de Souza Catojo, A. R., and Nunes, M. A. S. N. (2022). O desenvolvimento do pensamento computacional em alunos do ensino fundamental: Um mapeamento sistemático da literatura. *Simpósio Brasileiro de Informática na Educação (SBIE)*, pages 1324–1333.
10. Dienes, Z. P. (1967). Building up mathematics.
11. Filippa, D., Roque, G., and Pedrosa, S. (2018). Pesquisa-ação: possibilidades para a informática educativa. *Metodologia de Pesquisa Científica em Informática na Educação: Abordagem qualitativa de Pesquisa*, 3.
12. Grando, R. C. (2015). Recursos didáticos na Educação Matemática: jogos e materiais manipulativos. *Revista Eletrônica Debates em Educação Científica e Tecnológica*, 5(02), 393-416.
13. Kamii, C., Rummelsburg, J., and Kari, A. (2005). Teaching arithmetic to low-performing, low-ses first graders. *The journal of mathematical behavior*, 24(1):39–50.
14. Lima, T. F., Bairros, G., Franco, M. H. I., de Castro Bertagnolli, S., Peres, A., Okuyama, F. Y., and Nicolao, M. (2024). Explorando o pensamento criativo e computacional com aprendizagem baseada em problemas: Um relato de experiência com ScratchJr. In *Simpósio Brasileiro de Informática na Educação (SBIE)*, pages 3246–3254. SBC.
15. Lodi, M. and Martini, S. (2021). Computational thinking, between Papert and Wing. *Science & education*, 30(4):883–908.
16. Leite, E. P., Machado, T. C. L., Galiardi, N. S., & de Oliveira, W. A. A. (2024, November). Formação Pedagógica: o Material Dourado como Desenvolvimento de Ensino e Aprendizagem Matemática. In *Seminário de Educação (SemiEdu)* (pp. 514-519). SBC.
17. Papert, S. (1980). *Mindstorms: Children, computers, and powerful ideas*. New York: Basic Books.
18. Reis, A. Q. and Nehring, C. M. (2017). A contextualização no ensino de matemática: concepções e práticas. *Educação Matemática Pesquisa Revista do Programa de Estudos Pós Graduados em Educação Matemática*, 19(2).
19. Resnick, L. B. (1987). Education and learning to think.
20. Siedler, M. S., Cardoso, R. C., Souza, S. C., Schmidt, M. A., Tavares, T. A., and Primo, T. T. (2024). Aventuras digitais com material dourado: Jogo digital voltado ao aprendizado de matemática. In *Simpósio Brasileiro de Jogos e Entretenimento Digital (SB Games)*, pages 1453–1464. SBC.
21. Sociedade Brasileira de Pediatria (2024). Grupo de Trabalho e Saúde na Era Digital: Manual de Orientação. #Menos Telas #Mais Saúde. Sociedade Brasileira de Pediatria, Rio de Janeiro.
22. Souza, S. C., Cardoso, R. C., Tavares, T. A., Primo, T. T., and Siedler, M. S. (2024). Material dourado digital: Uma alternativa para o ensino de matemática na educação primária. In *Simpósio*

Brasileiro de Jogos e Entretenimento Digital (SBGames), pages 356–360. SBC.

23. Valente, J. A. (1997). Informática na educação: instrucionismo x construcionismo. *Manuscrito não publicado, NIED: UNICAMP*.
24. Wing, J. M. (2006). Computational thinking. *Communications of the ACM*, 49(3):33–35.
25. Xavier, E. A., Foss, L., da Costa Cavalheiro, S. A., da Silveira Soares, M. A., and Romio, L. C. (2021). Pensamento computacional integrado à matemática na BNCC: proposta para o primeiro ano do ensino fundamental. In *Simpósio Brasileiro de Informática na Educação (SBIE)*, pages 989–1001. SBC.